Comparing AI Agents in a Simplified RISK Game

Dissertation Project – MSc Financial Technology

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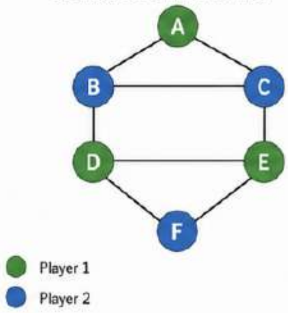
1. Research Aim

- Build and compare AI agents for a simplified RISK game.
- Test whether smarter decision-making improves performance.
- Measure both game success and computational cost.

Does increasing agent intelligence and computational complexity improve performance in a simplified RISK game?

2. Game Environment

- 2 players, 6 territories, dice-based combat.
- Territory ownership, reinforcement, attack, capture.
- Maximum 100 turns per game.
- 3,000 total games with fixed random seeds.
- 6 pairings, 500 games per pairing.



3. AI Agents

Random Agent: chooses moves randomly; baseline.

Rule-Based Agent: attacks using fixed strategic rules.

Heuristic Agent: scores actions using handcrafted heuristics.

Monte Carlo Agent: simulates battles to estimate best attack.

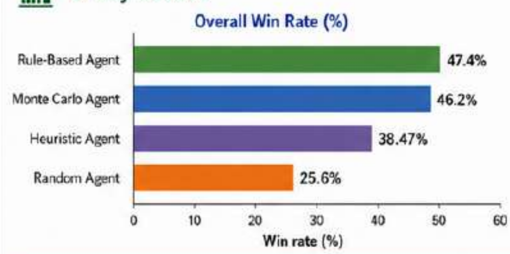
4. Methodology

- Agents alternated between Player 1 and Player 2.
- Metrics: win rate, draw rate, attack success, territories captured, armies lost, decision time, simulations.
- Automated tests and reproducible seeds were used.

7. Conclusion

- Smarter agents generally outperformed the random baseline.
- However, greater computational complexity did not guarantee the best overall result.
- Rule-based decision-making offered the best balance of performance and efficiency in this simplified environment.
- Future work: Monte Carlo Tree Search, full 42-territory map, and more advanced evaluation.

5. Key Results



Agent	Win rate	Draw rate	Attack success	Avg decision time	Avg simulations
Rule-Based Agent	47.4%	16.07%	49.55%	0.0051 ms	0
Monte Carlo Agent	46.2%	16.2%	48.33%	0.7492 ms	4572.6
Heuristic Agent	38.47%	14.0%	45.62%	0.0056 ms	0
Random Agent	25.6%	38.4%	55.32%	0.0043 ms	0

6. Main Findings

- Rule-Based Agent achieved the highest overall win rate.
- Monte Carlo Agent was a close second, but required far more computation.
- Random Agent had the lowest win rate and the highest draw rate.
- Monte Carlo averaged 4572.6 simulations per game.
- Player 1 had an advantage: 45.43% wins vs 33.4% for Player 2.
- Strongest head-to-head result: Monte Carlo beat Heuristic 58.8% to 38.4%.